# Harpreet Singh


**Research Engineer / Applied Scientist**

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## Summary


Research engineer and PhD student with 13+ years spanning embedded systems, applied AI, computational neuroscience, cybersecurity research, FPGA acceleration, and biomedical data acquisition. Experienced at turning research questions into reproducible software/hardware pipelines, labelled datasets, validation plans, experiment artifacts, metrics, publications, and working prototypes across unfamiliar technical domains.


## Core Skills

- Research engineering, experimental design, reproducible workflows, validation plans, metrics, documentation, technical writing
- Python, NumPy, Pandas, scikit-learn, TensorFlow, PyTorch, Matplotlib, Bayesian Networks, Markov Networks, NLP transformers
- Computational neuroscience, EEG/fNIRS, BCI, rodent raster dynamics, SNN, QSNN, QiSNN, adaptive early-exit inference
- Embedded systems, Raspberry Pi, STM32, C, firmware, data acquisition, sensors, SPI, I2C, UART, CAN, Ethernet
- FPGA acceleration, Vitis HLS, Vivado, MicroBlaze, Zynq, BRAM, AXI4-Lite, UART diagnostics, resource/timing evidence

## Professional Experience

**Research Assistant, Core-Hub Neuroengineering Solutions** | *University of Lethbridge | Apr 2025 - Present*

- Build current PhD neuroengineering research stack for EEG/fNIRS acquisition, CGX/XDF dataset creation, live BCI classifier/game control, embedded device control, and FPGA temporal QiSNN acceleration.
- Developed an 8-channel EEG + 8-channel fNIRS Raspberry Pi platform using ADS1299, ADS8688, DAC8565, SPI/GPIO, TDM MUX sequencing, live plotting, and CSV logging.
- Built CGX/Sheeg workflows that launch CGX Acquisition and LabRecorder, record labelled visual-stimulus XDF sessions, preprocess EEG into CSV datasets, and export model artifacts for live classification.
- Developed closed-loop BCI/FPGA and rodent-decision pipelines using ADS1299 frames, MicroBlaze, BRAM windows, temporal QiSNN/SNN accelerators, UDP/UART intent packets, and five-state game control.

**Research Assistant** | *University of Regina | Jan 2021 - Apr 2024*

- Researched uncertain-reasoning intrusion detection for DoS/DDoS detection using Bayesian Networks, Markov Networks, Zeek, Wireshark, feature engineering, and Python ML workflows.
- Published SECRYPT 2024 work on uncertain-reasoning IDS methods and presented findings to an international cybersecurity research audience.
- Built ML prototypes including a Bayesian liver-disorder diagnostic model with 85% accuracy and Punjabi BERT/NLP classifiers for 100K+ news articles with 92% classification accuracy.

**Independent Embedded Systems Developer** | *Self-Employed | Jun 2024 - Mar 2025*

- Designed a custom STM32F405RGT6 MicroPython development board, including multilayer PCB design, firmware flashing, sensor/actuator interfaces, power management, and hardware/software debugging.

**Senior Software Engineer** | *Planetcast Media Services Ltd. | Dec 2018 - Nov 2020*

- Led a 15-member team delivering embedded hardware and firmware for broadcast/media systems, including a network-controlled 10-channel video switcher and Ethernet temperature acquisition platform.
- Designed schematics and PCBs, selected components, collaborated on cabinet/mechanical integration, and developed C firmware with Ethernet, HTTP/LWIP, alarm, and control interfaces.

**Embedded Software Engineer / R&D Engineer** | *Spark Eighteen Pvt. Ltd.; Exicom Tele-systems Ltd. | Feb 2016 - Dec 2018*

- Developed STM32/AVR embedded systems for LiFi control, telecom power plants, DC interface cards, and automatic test equipment.
- Designed 15-20 W isolated SMPS sections with 78-81% measured efficiency and performed board bring-up across CAN, SPI Flash, I2C EEPROM, Ethernet/LWIP, and GUI test automation.

## Selected Projects

**Current PhD BCI Research Stack**

- Built an end-to-end research workflow spanning CGX EEG acquisition, five-state visual-stimulus labels, LabRecorder/XDF recording, preprocessing manifests, held-session model evaluation, quantized model export, live classifier replay, and FPGA/game-loop integration.

**QSNN / SNN / ANN FPGA Research**

- Defined hardware-aware research methodology comparing ANN, SNN, fixed-step QSNN, and adaptive early-exit QSNN on accuracy, latency, power/energy, FPGA utilization, and robustness.

**Rodent Decision Temporal QiSNN Pipeline**

- Transformed neural population raster dynamics into temporal fixed-point features for FPGA-oriented no-lick, left-lick, and right-lick decision prediction, with DANDI sample support and documented raster-to-time-series conversion.

## Publication

- Harpreet Singh et al., "An Uncertain Reasoning-Based Intrusion Detection System for DoS/DDoS Detection," SECRYPT 2024, ISBN 978-989-758-709-2, ISSN 2184-7711, pp. 771-776.
- Published thesis: Singh H. A robust intrusion detection system utilizing uncertain reasoning techniques in artificial intelligence. The University of Regina (Canada); 2024.

## Education

- PhD in Computational Behavioural Neuroscience, University of Lethbridge - In progress
- M.Sc. Computer Science, University of Regina - 2024
- Bachelor's equivalent in Electronics & Telecommunication Engineering, AMIETE/IETE - 2012